Jerry Wu

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Summary

Statistics PhD Student at the University of Toronto with a strong foundation in Economics and Computer Science. Passionate about uncovering the immense value within large, complex datasets. Highly skilled in writing robust code to extract key performance metrics, streamline data processing, and produce visually appealing, actionable visualizations.

Education

University of Toronto, St. George — *Toronto, Canada*
Docter of Philosophy - Statistical Sciences — *Sep. 2024 - Present*
- Relevant Courses: Methods of Applied Statistics, Theory and Methods for Complex Spatial Data, Time Series Analysis, Advanced Computational Methods for Statistics, Advanced Theory of Statistics.

University of Toronto, St. George — *Toronto, Canada*
Bachelors of Science
Statistics and Economics Double Majors (Data Analytics Focus), Computer Science Minor — *Sep. 2019 - May. 2023*
- CGPA: 3.75/4.00
- Relevant Courses: Methods of Data Analysis, Statistical Machine Learning, Economics with Machine Learning and Data Mining, Big-Data Tools for Economists, Introduction to Databases, Calculus, Linear Algebra.

Work Experience

University of Toronto — *Toronto, Canada*
Research Assistant (w. Doctor Meredith Franklin @ University of Toronto) — *Sep. 2022 - Present*
- Provided programming assistance using R to perform data cleaning, visualization, modelling.
- Collaborated in composing the journal articles for the first two projects described below.

University of Toronto — *Toronto, Canada*
Teaching Assistant — *Jan. 2023 - Present*
- Courses: An Introduction to Statistical Reasoning and Data Science, Probability, Statistics and Data Analysis, Statistical Methods for Machine Learning, Statistics Aid Centre
- Conducted weekly in-person tutorials for a group of 25 students, facilitating interactive discussions, reviewing course materials, and fostering a positive learning environment for understanding statistics.
- Provided ad-hoc support to undergraduate students by clarifying complex statistical concepts, guiding assignment problem-solving, and facilitating exam preparation across diverse statistics coursework.
- Evaluated and provided constructive feedback on student assignments to promote academic growth and development.

TMX Group — *Toronto, Canada*
Data Engineer (Intern) — *May. 2022 - Aug. 2022*
- Developed and implemented a robust fund fact sheet parser using Python and REGEX to extract critical information from PDF, TXT, and XML files stored on AWS S3. Designed the solution to be flexible and easily extendable for future variations.
- Actively participated in collaborative data quality assurance efforts, working closely with colleagues to identify, report, and resolve issues. Utilized a range of tools including DBeaver, GitHub, and Excel to ensure data accuracy.
- Wrote SQL scripts on Zeppelin, Redash, and DBeaver for ad-hoc data requests and QA tasks, utilizing PostgreSQL, MySQL.

Project Experience

Characterizing Unconventional Oil & Gas Development in the Permian Basin — *Toronto, Canada*
Research Assistant (w. Doctor Meredith Franklin @ University of Toronto) — *Sep. 2022 - Present*
- Automated data ETL process for VIIRS night flare data and Enverus data using R for U.S. oil and wells.
- Employed k-cluster, nearest neighbor, and regression techniques to map flare and well data, characterizing flaring activity in UOG wells. Variable selection and model optimization were performed for accuracy.

Modelling Hydrogen Sulfide Concentrations and Leaks from Oil Refineries in Los Angeles — *Toronto, Canada*
Research Assistant (w. Doctor Meredith Franklin @ University of Toronto) — *Feb. 2023 - Present*
- Conducted data cleaning, validation, and mapping on large data sets and shape files using R.
- Utilized GAM and XGBoost models with 80/20 split, 10-K CV, and box-cox transformations to achieve accurate predictions.
- Produced meaningful visualisations to illustrate the impact of the leakage and predicted H2S concentration in Los Angeles.
- Explored Extreme Value Theory distributions to accurately model pollutant concentrations with extreme values.

Understanding spatial characteristics of parking violations in Downtown Toronto *Toronto, Canada*

COURSE - THEORY AND METHODS FOR COMPLEX SPATIAL DATA *Oct. 2025 - Dec. 2025*

- Analyzed roughly 1.1 million parking tickets issued in Downtown Toronto across 2023 and 2024 to identify spatial clustering and temporal trends.
- Fitted multiple point pattern models, including Inhomogeneous Poisson Process (IPP) models and Log-Gaussian Cox Process (LGCP) models with Kernel Density Estimation covariates, to estimate violation intensities based on local features.
- Leveraged HDBSCAN to identify significant violation hotspots, revealing strong clustering near major intersections and areas with high population density.

Predicting traffic accident severity and the effect of Cannabis legalization *Toronto, Canada*

COURSE - ECONOMICS WITH MACHINE LEARNING AND DATA MINING *Feb. 2023 - Apr. 2023*

- Utilized regression, random forests, nearest neighbor clustering, and neural network algorithms to predict traffic severity rate in Quebec, Canada.
- Cleaned and processed data from the Quebec Data Partnership using Python.
- Performed model optimization through hyperparameter tuning and conducted model validation.

Publications

Modeling community hydrogen sulfide exposure in an urban industrial area during routine and extreme events *Environmental Pollution*

Meredith Franklin, **Jerry Yuxuan Wu**, Brandyn Ruiz, Jill Johnston, Arbor J.L. Quist *May 2026*

Source Contributions to Air Pollution in the Permian Basin: Evaluating Emissions from Unconventional Oil and Gas Activities *Environmental Science and Technology Air*

Meredith Franklin*, Gunnar Schade*, Dani Caputi, Lara J. Cushing, Ryan W. Daly, Gabriel Greenberg, Detlev Helmig, Jacques Hueber, Jill E. Johnston, Katherine E. Potter, Michel Stahli, **Jerry Yuxuan Wu** *Apr. 2026*

Airborne Radioactivity in the New Mexico Oil and Gas Development area of the Permian Basin *Elementa: Science of the Anthropocene*

Dani Caputi, Meredith Franklin, Gabriel Greenberg, Jacques Hueber, Jill Johnston, Kat Potter, Gunnar Schade, Michel Stahli, **Jerry Yuxuan Wu**, Lara Cushing, Ryan Daly, Detlev Helmig *Aug. 2026*

Presentation

Reading Course in Statistical Sciences Presentation *Toronto, Canada*

PRESENTER CHARACTERIZING UNCONVENTIONAL OIL AND GAS DEVELOPMENT IN THE PERMIAN BASIN *Dec. 2022*

- Presented the context and findings in the project with fellow peers and professors.

Honors & Awards

2024 - 2026 **Queen Elizabeth II Graduate Scholarship in Science & Technology**, *Toronto, Canada*

2024 **Department of Statistics Graduate Fellowship**, *Toronto, Canada*

2021 - 2023 **Dean's List Scholar**, *Toronto, Canada*

2021 **In-course scholarship**, *Toronto, Canada*

Extracurricular Activity

University of Toronto Application Development Association *Toronto, Canada*

WRITER IN MARKETING DEPARTMENT *Sep. 2019 - 2023*

- Writer for the social media account, to provide information, tutorial, news about university life and computer science.
- Helped to organize and prepare events hosted by UTADA, such as IF2020

Skills

Programming Languages R, Python, SQL, Stata, Presto, Java, Javascript, Latex, HTML, CSS

Libraries R: Tidyverse, MASS, olrss, lme4, openair. Python: Pandas, Matplotlib, Scikit-learn AWS: S3, CLI, Boto3

Other Tools OpenAI API, Power BI, Git, Jupyter, DBeaver, Google Workspace, Microsoft Office, Slack